Partial Report Composition Preview

Layout diagnostic only: selected numerical sections, not the complete final report or publication acceptance.

Selected Numerical Sections

Conclusions and Geometric Mechanism Synthesis

The following descriptive ranges and interval-direction counts cover only the eight selected zero-lean conditions per club (curve ×1/×2, shaft-twist SD 1°/2°, shaft-only/combined2). Positive SG favors the named curve over Straight. Ranges summarize cell point estimates, not pooled confidence intervals. Shared draws make these cells dependent; interval directions are pointwise Monte Carlo statements, not empirical or multiplicity-adjusted discoveries.

Club	Pattern	Selected SG Estimate Range (Strokes)	Positive / Negative / Zero-Spanning 95% Intervals	Common-Three Scored n Range / 10,000
driver	Draw	+2.691e-05 to +0.0078	6 / 0 / 2	10,000–10,000 / 10,000
driver	Fade	-0.00146 to +2.344e-04	1 / 4 / 3	10,000–10,000 / 10,000
seven_iron	Draw	-0.001356 to -2.883e-04	0 / 8 / 0	9,996–10,000 / 10,000
seven_iron	Fade	+8.934e-04 to +0.00258	8 / 0 / 0	9,996–10,000 / 10,000
pitching_wedge	Draw	+9.508e-05 to +0.01112	6 / 0 / 2	9,999–10,000 / 10,000
pitching_wedge	Fade	-0.01097 to +2.919e-04	1 / 6 / 1	9,999–10,000 / 10,000

The relative scoring pattern depends on club and delivery-noise assumptions; these selected rows do not establish a universal best shape. An interval spanning zero does not establish equivalence. The direct paired interaction tables above answer SD 1→2 and curve ×1→×2 on the same six-population intersection; their intervals must not be obtained by subtracting the marginal intervals summarized here.

For the Straight, curve ×1, shaft-only subset below, nominal targets are matched before noise. The realized downrange errors and correlations use each pattern's individual supported endpoints; SD changes need not preserve mean carry.

Club	Shaft-Twist SD (°)	Lateral/Downrange Correlation	Mean Downrange Error (m)	Individual Supported / Attempted
driver	1	+0.9788	-0.505	10,000 / 10,000
driver	2	+0.9186	-2.005	10,000 / 10,000
seven_iron	1	-0.9960	-0.055	10,000 / 10,000
seven_iron	2	-0.9841	-0.220	10,000 / 10,000
pitching_wedge	1	-0.9982	-0.023	10,000 / 10,000
pitching_wedge	2	-0.9930	-0.095	10,000 / 10,000

Positive lateral coordinates mean right. A positive lateral/downrange correlation associates rightward errors with longer downrange endpoints; a negative correlation associates leftward errors with longer endpoints. This describes the selected simulated population, not a causal face-angle attribution.

At zero shaft twist, delivered loft is L − lean and the rigid geometry gives d(loft)/d(face) = cot(lie) cos(L − lean)/cos(L), positive for these setups. Opening the face by shaft rotation locally increases delivered loft; closing locally decreases it. Fixed primitive twist SD approximately preserves the first-order loft SD across lean while face SD changes. Launch, speed, spin and aerodynamics determine whether that loft change increases carry. See the Scientific Design and the existing editable Shot-Pattern Reference.

Fixed-target controls remove nominal range mismatch, but include rematched face, path and speed and retain conditional support. No causal fraction of dispersion or scoring is assigned to distance, lean or curvature. Narrow Monte Carlo intervals exclude assumed head-property and contact-law errors and unmeasured golfer variability.

Resolved Zero-Lean Curve and Variability Comparison

This resolved selection retains original primary accounting and explicitly identifies the four PW strong-curve face-bounds supplements. Each condition key below maps to one exact cell and protocol. Positive paired SG favors the named curve over Straight; intervals describe Monte Carlo sampling. SG is per identical hypothetical opportunity, not per round.

Condition Key

C01: `driver_lean0_curve1_rotation1_shaft_only`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C02: `driver_lean0_curve1_rotation1_combined2`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C03: `driver_lean0_curve1_rotation2_shaft_only`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C04: `driver_lean0_curve1_rotation2_combined2`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C05: `driver_lean0_curve2_rotation1_shaft_only`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×2.

C06: `driver_lean0_curve2_rotation1_combined2`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×2.

C07: `driver_lean0_curve2_rotation2_shaft_only`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×2.

C08: `driver_lean0_curve2_rotation2_combined2`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×2.

C09: `seven_iron_lean0_curve1_rotation1_shaft_only`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C10: `seven_iron_lean0_curve1_rotation1_combined2`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C11: `seven_iron_lean0_curve1_rotation2_shaft_only`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C12: `seven_iron_lean0_curve1_rotation2_combined2`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C13: `seven_iron_lean0_curve2_rotation1_shaft_only`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×2.

C14: `seven_iron_lean0_curve2_rotation1_combined2`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×2.

C15: `seven_iron_lean0_curve2_rotation2_shaft_only`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×2.

C16: `seven_iron_lean0_curve2_rotation2_combined2`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×2.

C17: `pitching_wedge_lean0_curve1_rotation1_shaft_only`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C18: `pitching_wedge_lean0_curve1_rotation1_combined2`; protocol `original_primary/1`. Shaft twist SD 1°; curve ×1.

C19: `pitching_wedge_lean0_curve1_rotation2_shaft_only`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C20: `pitching_wedge_lean0_curve1_rotation2_combined2`; protocol `original_primary/1`. Shaft twist SD 2°; curve ×1.

C21: `pitching_wedge_lean0_curve2_rotation1_shaft_only`; protocol `pw-primary-face-bounds/1`. Shaft twist SD 1°; curve ×2.

C22: `pitching_wedge_lean0_curve2_rotation1_combined2`; protocol `pw-primary-face-bounds/1`. Shaft twist SD 1°; curve ×2.

C23: `pitching_wedge_lean0_curve2_rotation2_shaft_only`; protocol `pw-primary-face-bounds/1`. Shaft twist SD 2°; curve ×2.

C24: `pitching_wedge_lean0_curve2_rotation2_combined2`; protocol `pw-primary-face-bounds/1`. Shaft twist SD 2°; curve ×2.

Paired Course Scoring

Condition	Common-Three Scored / Attempted	Draw Δ SG [95%]	Fade Δ SG [95%]
C01	10,000 / 10,000	+7.462e-05 [-7.104e-05, +2.190e-04]	+6.826e-05 [-5.228e-05, +1.705e-04]
C02	10,000 / 10,000	+2.691e-05 [-1.642e-04, +2.204e-04]	+4.218e-05 [-1.739e-04, +2.646e-04]
C03	10,000 / 10,000	+0.00289 [+0.00246, +0.00338]	-0.001225 [-0.00162, -8.629e-04]
C04	10,000 / 10,000	+0.002195 [+0.001695, +0.002636]	-9.277e-04 [-0.001382, -5.147e-04]
C05	10,000 / 10,000	+3.250e-04 [+1.042e-04, +5.513e-04]	+2.344e-04 [+5.011e-05, +4.290e-04]
C06	10,000 / 10,000	+5.626e-04 [+2.392e-04, +8.978e-04]	+9.462e-05 [-2.218e-04, +4.419e-04]
C07	10,000 / 10,000	+0.0078 [+0.00713, +0.00852]	-0.00146 [-0.001974, -9.630e-04]
C08	10,000 / 10,000	+0.00668 [+0.00594, +0.00744]	-8.347e-04 [-0.001478, -2.621e-04]
C09	10,000 / 10,000	-5.056e-04 [-5.137e-04, -4.968e-04]	+0.001047 [+0.0010332, +0.001059]
C10	9,996 / 10,000	-7.450e-04 [-8.996e-04, -5.795e-04]	+8.934e-04 [+7.090e-04, +0.001078]
C11	10,000 / 10,000	-4.992e-04 [-5.776e-04, -4.544e-04]	+0.001067 [+9.861e-04, +0.001182]

Condition	Common-Three Scored / Attempted	Draw Δ SG [95%]	Fade Δ SG [95%]
C12	9,996 / 10,000	-8.053e-04 [-0.001038, -5.918e-04]	+0.00111 [+8.442e-04, +0.001355]
C13	10,000 / 10,000	-3.978e-04 [-4.106e-04, -3.851e-04]	+0.0025526 [+0.00252, +0.00258]
C14	9,996 / 10,000	-0.001356 [-0.00165, -0.001025]	+0.001975 [+0.001623, +0.002286]
C15	10,000 / 10,000	-2.883e-04 [-4.426e-04, -1.306e-04]	+0.002582 [+0.002453, +0.002736]
C16	9,996 / 10,000	-0.001053 [-0.001428, -6.655e-04]	+0.002294 [+0.00193, +0.002645]
C17	10,000 / 10,000	+0.004434 [+0.004363, +0.004502]	-0.00445 [-0.004517, -0.00438]
C18	9,999 / 10,000	+9.508e-05 [-6.221e-05, +2.800e-04]	+9.390e-05 [-3.145e-05, +2.411e-04]
C19	10,000 / 10,000	+0.005527 [+0.005466, +0.005576]	-0.005487 [-0.005537, -0.005427]
C20	9,999 / 10,000	+4.844e-04 [+3.075e-04, +6.523e-04]	-3.337e-04 [-5.016e-04, -1.721e-04]
C21	10,000 / 10,000	+0.008878 [+0.008738, +0.009015]	-0.008934 [-0.009068, -0.008798]
C22	9,999 / 10,000	+2.513e-04 [-3.605e-05, +5.326e-04]	+2.919e-04 [+3.620e-05, +5.922e-04]
C23	10,000 / 10,000	+0.011117 [+0.010994, +0.01122]	-0.010966 [-0.011068, -0.010844]
C24	9,999 / 10,000	+0.001046 [+7.406e-04, +0.001332]	-6.272e-04 [-9.681e-04, -2.936e-04]

Individual Supported Endpoints

These individual populations can exceed the common-three scored intersection. Each entry is supported / attempted; no unsupported shots are retried.

Condition	Straight	Draw	Fade
C01	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C02	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C03	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C04	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C05	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C06	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C07	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C08	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C09	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000

Condition	Straight	Draw	Fade
C10	9,996 / 10,000	9,996 / 10,000	9,996 / 10,000
C11	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C12	9,996 / 10,000	9,996 / 10,000	9,996 / 10,000
C13	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C14	9,996 / 10,000	9,996 / 10,000	9,996 / 10,000
C15	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C16	9,996 / 10,000	9,996 / 10,000	9,996 / 10,000
C17	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C18	9,999 / 10,000	9,999 / 10,000	9,999 / 10,000
C19	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C20	9,999 / 10,000	9,999 / 10,000	9,999 / 10,000
C21	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C22	9,999 / 10,000	9,999 / 10,000	9,999 / 10,000
C23	10,000 / 10,000	10,000 / 10,000	10,000 / 10,000
C24	9,999 / 10,000	9,999 / 10,000	9,999 / 10,000

Ordered Contrast Key

E01 — shaft_rotation_sd:

A: `original_control/1; driver_lean0_curve1_rotation1_shaft_only`.

B: `original_control/1; driver_lean0_curve1_rotation2_shaft_only`.

E02 — curve_strength:

A: `original_control/1; driver_lean0_curve1_rotation2_shaft_only`.

B: `original_control/1; driver_lean0_curve2_rotation2_shaft_only`.

E03 — shaft_rotation_sd:

A: `original_control/1; driver_lean0_curve1_rotation1_combined2`.

B: `original_control/1; driver_lean0_curve1_rotation2_combined2`.

E04 — curve_strength:

A: `original_control/1; driver_lean0_curve1_rotation2_combined2`.

B: `original_control/1; driver_lean0_curve2_rotation2_combined2`.

E05 — shaft_rotation_sd:

A: `original_control/1; seven_iron_lean0_curve1_rotation1_shaft_only`.

B: `original_control/1; seven_iron_lean0_curve1_rotation2_shaft_only`.

E06 — curve_strength:

A: `original_control/1; seven_iron_lean0_curve1_rotation2_shaft_only`.

B: `original_control/1`; `seven_iron_lean0_curve2_rotation2_shaft_only`.

E07 — shaft_rotation_sd:

A: `original_control/1`; `seven_iron_lean0_curve1_rotation1_combined2`.

B: `original_control/1`; `seven_iron_lean0_curve1_rotation2_combined2`.

E08 — curve_strength:

A: `original_control/1`; `seven_iron_lean0_curve1_rotation2_combined2`.

B: `original_control/1`; `seven_iron_lean0_curve2_rotation2_combined2`.

E09 — shaft_rotation_sd:

A: `original_control/1`; `pitching_wedge_lean0_curve1_rotation1_shaft_only`.

B: `original_control/1`; `pitching_wedge_lean0_curve1_rotation2_shaft_only`.

E10 — curve_strength:

A: `original_control/1`; `pitching_wedge_lean0_curve1_rotation2_shaft_only`.

B: `pw-primary-face-bounds/1`; `pitching_wedge_lean0_curve2_rotation2_shaft_only`.

E11 — shaft_rotation_sd:

A: `original_control/1`; `pitching_wedge_lean0_curve1_rotation1_combined2`.

B: `original_control/1`; `pitching_wedge_lean0_curve1_rotation2_combined2`.

E12 — curve_strength:

A: `original_control/1`; `pitching_wedge_lean0_curve1_rotation2_combined2`.

B: `pw-primary-face-bounds/1`; `pitching_wedge_lean0_curve2_rotation2_combined2`.

Contrast	Curved Pattern	B − A Relative SG [95% MC]	Common Six / Attempted
E01	Draw	+0.00282 [+0.002346, +0.00329]	10,000 / 10,000
E01	Fade	-0.001293 [-0.001708, -8.956e-04]	10,000 / 10,000
E02	Draw	+0.004905 [+0.00438, +0.00545]	10,000 / 10,000
E02	Fade	-2.354e-04 [-5.740e-04, +1.124e-04]	10,000 / 10,000
E03	Draw	+0.00217 [+0.0016, +0.002706]	10,000 / 10,000
E03	Fade	-9.699e-04 [-0.001502, -5.028e-04]	10,000 / 10,000
E04	Draw	+0.00449 [+0.00394, +0.00507]	10,000 / 10,000
E04	Fade	+9.299e-05 [-2.735e-04, +4.642e-04]	10,000 / 10,000
E05	Draw	+6.449e-06 [-7.089e-05, +4.934e-05]	10,000 / 10,000

Contrast	Curved Pattern	B − A Relative SG [95% MC]	Common Six / Attempted
E05	Fade	+2.000e-05 [-5.979e-05, +1.358e-04]	10,000 / 10,000
E06	Draw	+2.109e-04 [+9.164e-05, +3.578e-04]	10,000 / 10,000
E06	Fade	+0.001515 [+0.001433, +0.001629]	10,000 / 10,000
E07	Draw	-6.028e-05 [-2.966e-04, +1.540e-04]	9,996 / 10,000
E07	Fade	+2.175e-04 [-7.965e-05, +5.113e-04]	9,996 / 10,000
E08	Draw	-2.476e-04 [-4.948e-04, +2.780e-05]	9,996 / 10,000
E08	Fade	+0.001183 [+9.817e-04, +0.001365]	9,996 / 10,000
E09	Draw	+0.001093 [+0.001026, +0.001164]	10,000 / 10,000
E09	Fade	-0.001037 [-0.001105, -9.711e-04]	10,000 / 10,000
E10	Draw	+0.00559 [+0.005529, +0.005643]	10,000 / 10,000
E10	Fade	-0.005479 [-0.00553, -0.005418]	10,000 / 10,000
E11	Draw	+3.893e-04 [+2.185e-04, +5.506e-04]	9,999 / 10,000
E11	Fade	-4.276e-04 [-5.452e-04, -3.150e-04]	9,999 / 10,000
E12	Draw	+5.621e-04 [+4.164e-04, +7.167e-04]	9,999 / 10,000
E12	Fade	-2.935e-04 [-4.911e-04, -8.723e-05]	9,999 / 10,000

B − A is the change in (curved − Straight) SG on the same six-population supported-index intersection, with A and B identified above. It is not a causal distance contribution. Intervals are pointwise Monte Carlo sampling intervals; shared seeded errors correlate cells and do not validate player performance.